Fitness analysis

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1 Setup

Analysis of the fitness of selected *Vicia faba* genotypes infested or not with *A. fabae* AF001 or *A. pisum* P123amp aphids.

See the data source.

1.1 library

1.2 Data

2 Summaries for uninfested plants

2.1 Total of seeds per genotype in control conditions

name	total_n_seeds
IG 11734	568
CANAKKALE	721
M10868	1403
CS 20 DK	964

> **Note**
>
> In control conditions, the total number of seeds for 7 to 9 plants per genotype ranges from 568 to 1403.

2.2 How many seeds per genotype taking the number of plants into account (only control)?

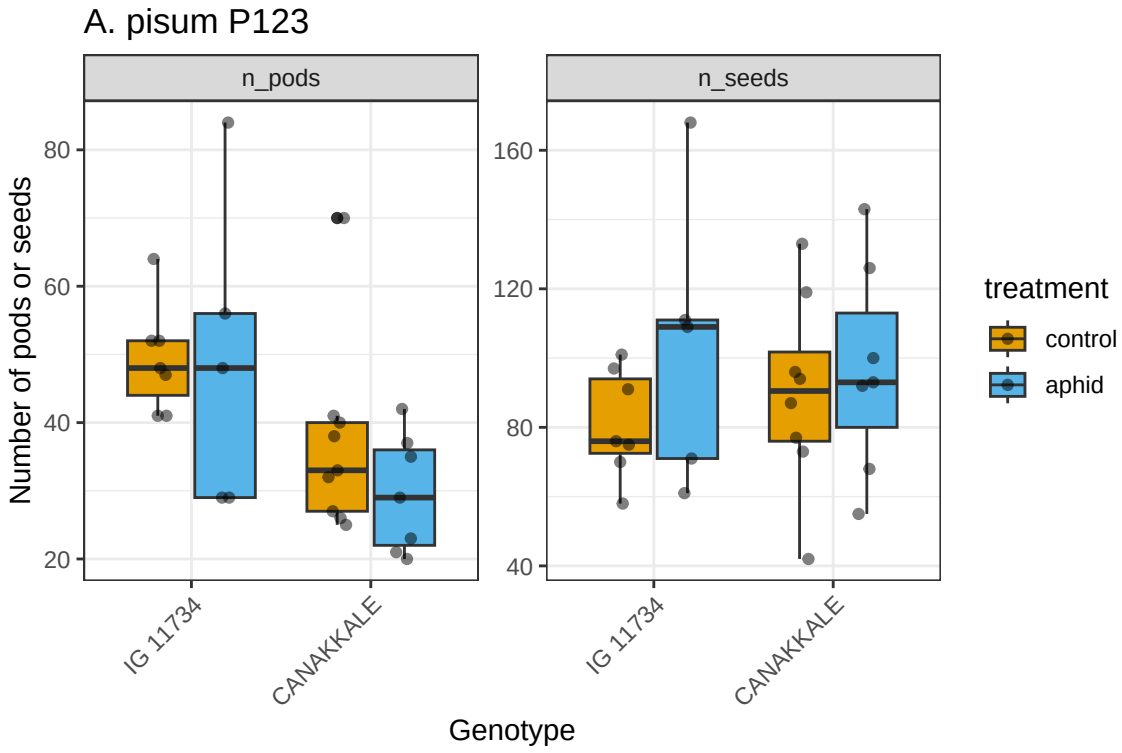
name	total_n_seeds	aphid_species	n_plants	n_seeds_per_plant
IG 11734	568	A. pisum P123	7	81
CANAKKALE	721	A. pisum P123	9	80
M10868	1403	A. fabae AF001	8	175
CS 20 DK	964	A. fabae AF001	9	107

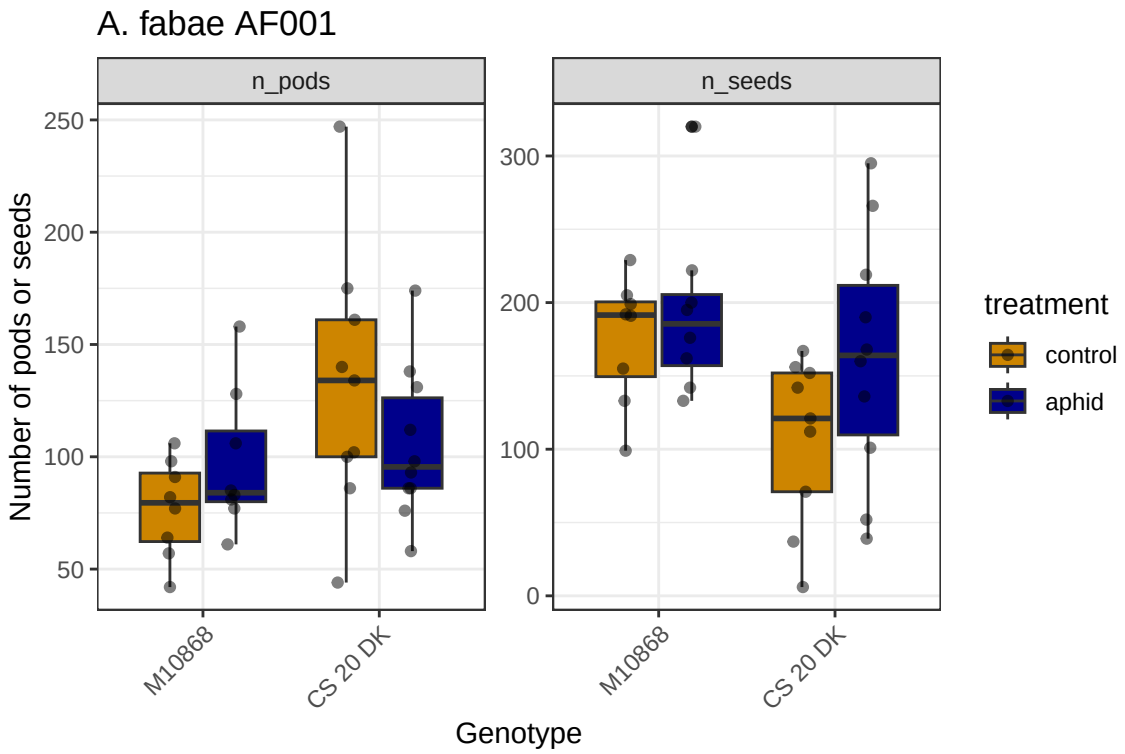
> **Note**
>
> In control conditions, the average number of seeds per plant ranges from 80 to 175.

3 Aphid effect on fitness

3.1 Aphid effect on number of pods and seeds

3.1.1 Plots of n_seeds = f(genotype, aphid treatment)





3.1.2 Statistical tests n_seeds/pods = f(genotype, aphid treatment)

ANOVA for the effect of A. pisum P123 on the number of pods.

term	df	sumsq	meansq	statistic	p.value	significance
treatment	1	142.74107	142.74107	0.7752696	0.3873241	ns
name	1	1641.22853	1641.22853	8.9140042	0.0064211	**
treatment:name	1	87.62722	87.62722	0.4759297	0.4968891	ns
Residuals	24	4418.83175	184.11799	NA	NA	ns

ANOVA for the effect of A. pisum P123 on the number of seeds.

term	df	sumsq	meansq	statistic	p.value	significance
treatment	1	1272.66852	1272.66852	1.4859200	0.2352056	ns
name	1	22.68876	22.68876	0.0264905	0.8721296	ns
treatment:name	1	433.33386	433.33386	0.5059443	0.4840519	ns
Residuals	23	19699.16071	856.48525	NA	NA	ns

! Important

For *A. pisum* P123:
No effect on aphid treatment on the number of pods or number of seeds. Variability is high between plants. The number of pods is significantly different between genotypes, but not the number of seeds.

ANOVA for the effect of A. fabae AF001 on the number of pods.

term	df	sumsq	meansq	statistic	p.value	significance
treatment	1	178.0729	178.0729	0.113452	0.7385176	ns
name	1	8253.9290	8253.9290	5.258658	0.0287798	*
treatment:name	1	4823.5021	4823.5021	3.073100	0.0894830	ns
Residuals	31	48657.2389	1569.5884	NA	NA	ns

ANOVA for the effect of A. fabae AF001 on the number of seeds.

term	df	sumsq	meansq	statistic	p.value	significance
treatment	1	12104.668	12104.668	2.9450669	0.0961192	ns
name	1	21061.618	21061.618	5.1242938	0.0307397	*
treatment:name	1	2987.221	2987.221	0.7267911	0.4004641	ns
Residuals	31	127414.664	4110.150	NA	NA	ns

! Important

For *A. fabae* AF001: Aphid treatment has no significant effect on the number of pods and number of seeds. The number of pods and seeds is significantly different between genotypes.